CO-CIRCULATION OF ALL 4 SEROTYPES OF DENGUE VIRUS IN ENDEMIC REGION OF SAURASHTRA, GUJARAT DURING 2019.

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ABSTRACT

Dengue is a fast emerging mosquito-borne viral disease endemic in many regions, causing a severe flu-like illness and sometimes potentially lethal complications. Its causative agent, the dengue-virus, has four serotypes ,DENV-1, DENV-2, DENV-3 and DENV-4.

To obtain an epidemiological update, 11,181 samples of clinically suspected cases were tested for positivity by ELISA and 120 out of the confirmed cases were serotyped by RT-PCR, in Saurashtra region of Gujarat,India, during the year 2019.

The results established endemicity of dengue with higher prevalence in male adult population and seasonal peaks in monsoon. All 4 serotypes of Dengue virus were found to be co-circulating in the region along with the evidence of co-infection by 2 serotypes in a single patient,hence resulting in variable clinical manifestations.

The lack of cross-protective immunity between heterologous serotypes raises concern about an impending global pandemic and can be prevented using the sole and most effective strategy of vector-control and personal protection with increased vigilance during monsoon.

KEYWORDS:- Dengue-virus, Co-circulation, Seasonal trends.

INTRODUCTION

- Dengue is common in >100 countries around the world and is a rapidly emerging threat for global pandemics. 40% of the world's population lives in areas with a risk of dengue.[1]
- Caused by: Dengue virus (DENV) [genus-flavivirus]
- Vector: Aedes aegypti mosquito (breeds in artificial water collections)
- Disease manifestations:
  - Dengue Fever(DF)
  - Dengue Hemorrhagic Fever (DHF)
  - Dengue Shock Syndrome (DSS).
- There are 4 antigenically different serotypes.

DENV-1 DENV-2 DENV-3 DENV-4.

(although there is report of 2013 that a fifth serotype has been found):[2]

- These may either mono-infect or co-infect a patient based on their prevalence and predominance in the given region and time, resulting in disease of varying severity, due to lack of cross-protection.
- Dengue is often a leading cause of illness in areas with risk.[1]

OBJECTIVES

To study:

- **Epidemiological update** of Dengue in Saurashtra region of Gujarat, India.
- **Circulation** of DENV serotypes in population.
- **Co-infection** patterns of the 4 serotypes and discuss its **prevalence**.
- **Effective strategy of control** from the results.

MATERIALS & METHODS

Total 11,181 serum samples of clinically suspected dengue cases (Jan 2019 to Dec 2019.)

Fever

<5 days → ELISA test for Dengue NS1-Antigen using Microlisa test kit

>5 days → Test for Dengue IgM Antibody using NIV Pune test kit

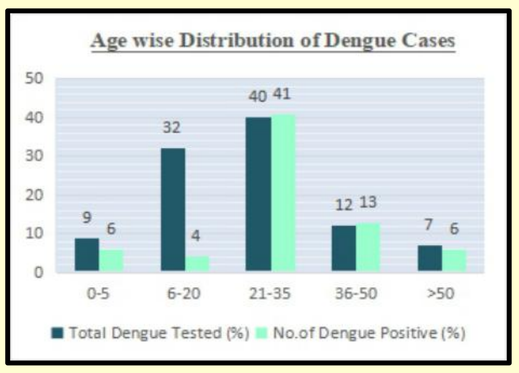
Serologically confirmed positive cases

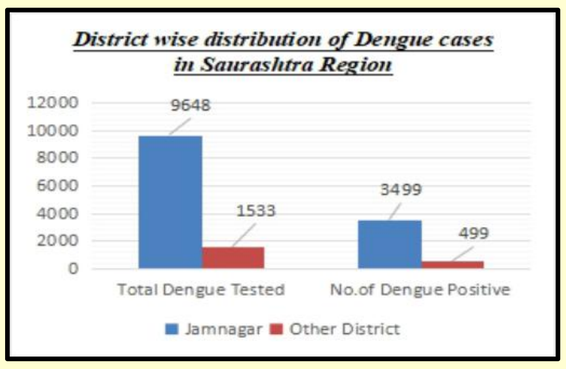
120 samples from different geographical areas of Saurashtra region selected

Serotyping by Dengue-specific RT-PCR

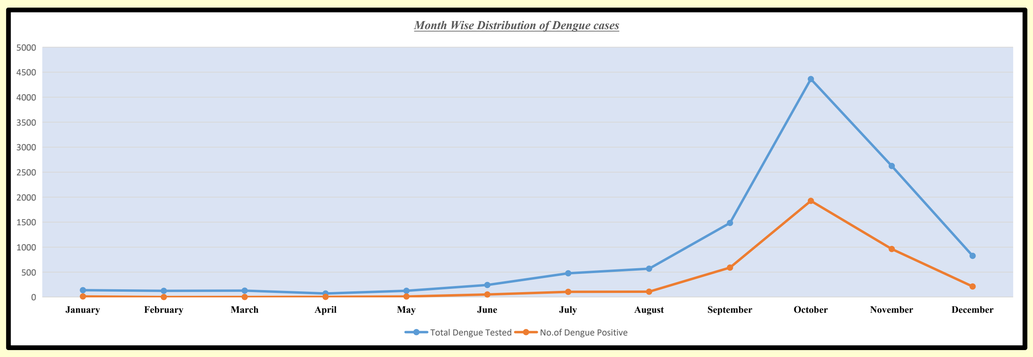
RESULTS

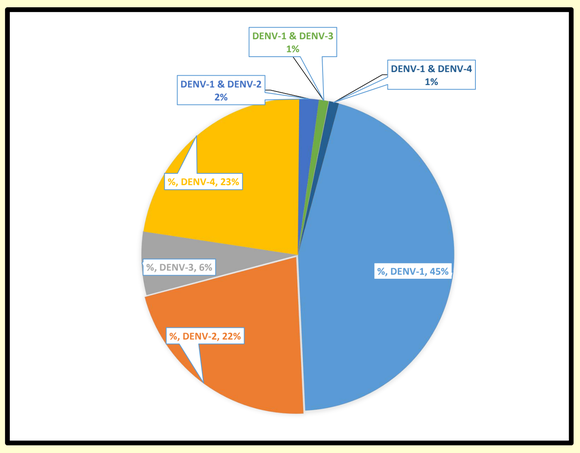
- Samples tested: 11,181
- Serologically confirmed positive cases : **3998 (36%)**
- **Male**-cases (60%) > Female-cases
- Most affected age-groups : **21-35 years (41%)**; 6-20 years (34%)





- Maximum cases recorded in the **Jamnagar** district of Saurashtra region.
- Seasonal trend:
  Gradual increase in positivity starting from June **(Monsoon)** , with a **peak in October** (39%)





- Co-circulation of **all four serotypes** observed.
- Predominance of **DENV-1 (44.8%)**, followed by DENV-4 (22.6%), DENV-2(21.5%) and DENV-3 (6.45%).
- Co- infections : **DENV-1 & DENV-2(2%)**, DENV-1 & DENV-3 and DENV-1(1%) & DENV-4(1%)

CONCLUSION

- ✓ Higher prevalence in **adult male population** indicates increased risk of infection **outdoors** .
- ✓ Seasonal peak during **monsoon and post-monsoon** period coincides with **increased vector breeding**.
- ✓ All 4 serotypes circulated with predominance of DENV-1 type in the Saurashtra region during 2019.
  - The four serotypes can co-circulate because the **immunity to one serotype does not confer protection from infection by a heterotopous serotype.**
  - This property to **co-circulate and co-infect** renders it a **global pandemic potential**.
- ✓ Hence, most effective strategy of control : Rigorous implementation of **vector-control measures** & **personal protection with increased vigilance during monsoon.**

REFERENCES

1.https://www.cdc.gov/
2 http://www.denguevirusnet.com/dengue-virus.html